

```
In [1]: !pip install torch torchvision torchaudio scikit-learn medmnist
```

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Requirement already satisfied: torch in /opt/conda/lib/python3.13/site-packages (2.13.0)
Requirement already satisfied: torchvision in /opt/conda/lib/python3.13/site-packages (0.28.0)
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Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.88)

Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.96)

Requirement already satisfied: nvidia-cufft==12.0.0.61.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (12.0.0.61)

Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.3.33)

Requirement already satisfied: nvidia-cufile==1.15.1.6.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (1.15.1.6)

Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.85)

Requirement already satisfied: nvidia-curand==10.4.0.35.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (10.4.0.35)

Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (12.0.4.66)

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Requirement already satisfied: nvidia-nvtx==13.0.85.* in /opt/conda/lib/python3.13/site-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch) (13.0.85)

Requirement already satisfied: cuda-pathfinder>=1.4.2 in /opt/conda/lib/python3.13/site-packages (from cuda-bindings<14,>=13.0.3->torch) (1.6.0)

Requirement already satisfied: numpy in /opt/conda/lib/python3.13/site-packages (from torchvision) (2.5.1)

Requirement already satisfied: pillow!>=8.3.*,>=5.3.0 in /opt/conda/lib/python3.13/site-packages (from torchvision) (12.3.0)

Requirement already satisfied: scipy>=1.10.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (1.18.0)

Requirement already satisfied: joblib>=1.4.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (1.5.3)

Requirement already satisfied: narwhals>=2.0.1 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (2.24.0)

Requirement already satisfied: threadpoolctl>=3.5.0 in /opt/conda/lib/python3.13/site-packages (from scikit-learn) (3.6.0)

Requirement already satisfied: pandas in /opt/conda/lib/python3.13/site-packages (from medmnist) (3.0.5)

Requirement already satisfied: scikit-image in /opt/conda/lib/python3.13/site-packages (from medmnist) (0.26.0)

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Requirement already satisfied: fire in /opt/conda/lib/python3.13/site-packages (from medmnist) (0.7.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/lib/python3.13/site-packages (from sympy>=1.13.3->torch) (1.3.0)

Requirement already satisfied: termcolor in /opt/conda/lib/python3.13/site-packages (from fire->medmnist) (3.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.13/site-packages (from jinja2->torch) (3.0.3)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.13/site-packages (from pandas->medmnist) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.13/site-packages (from python-dateutil>=2.8.2->pandas->medmnist) (1.17.0)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (2.37.4)

Requirement already satisfied: tifffile>=2022.8.12 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (2026.7.31)

Requirement already satisfied: packaging>=21 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (26.2)

Requirement already satisfied: lazy-loader>=0.4 in /opt/conda/lib/python3.13/site-packages (from scikit-image->medmnist) (0.5)

In [2]:

```
"""
Created on Mon Aug4 17:35:36 2025

@author: vigo
"""

import os
import sys
import argparse
import time
import pickle
from tqdm import trange
import numpy as np
import PIL
import copy
import random
import torch
import torch.nn as nn
import torch.nn.functional as F
import torch.optim as optim
import torch.utils.data as data
import torchvision
import torchvision.transforms as transforms
from torchvision.models import resnet18, resnet50
import matplotlib.pyplot as plt
from scipy.io import savemat

#Related medmnist dataset
import medmnist
from medmnist import INFO, Evaluator

# Check for Cuda
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```

seed = 27
torch.manual_seed(seed)
random.seed(seed)

# Hyper-parameters
n_epochs = 10
batch_size = 128
lr = 0.0005

#Network Hyper-parameters
hidden_size = 512

data_flag = 'organamnist'
info = INFO[data_flag]
task = info['task']
n_channels = info['n_channels']
n_classes = len(info['label'])
DataClass = getattr(medmnist, info['python_class'])

#To work with 28x28 size images
data_transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize(mean=[.5], std=[.5])])

### To work with 224x224 resolution images
#data_transform = transforms.Compose(
#    #[transforms.Resize((224, 224), interpolation=PIL.Image.NEAREST),
#    #transforms.ToTensor(),
#    #transforms.Normalize(mean=[0.5], std=[0.5])])

# Image Size 28x28. Set RGB =True if the dataset contains colour images
train_dataset = DataClass(split='train', transform=data_transform, download=True)
val_dataset = DataClass(split='val', transform=data_transform, download=True, as
test_dataset = DataClass(split='test', transform=data_transform, download=True,

# Loaders that give 64 example batches
train_loader = data.DataLoader(dataset=train_dataset,
                                batch_size=batch_size,
                                shuffle=True)

val_loader = data.DataLoader(dataset=val_dataset,
                              batch_size=batch_size,
                              shuffle=False)

test_loader = data.DataLoader(dataset=test_dataset,
                              batch_size=batch_size,
                              shuffle=False)

image, label = train_dataset[0] # Get one sample image
print(f"Image shape: {image.shape}")
input_size = image.shape[0] * image.shape[1] * image.shape[2]
print(f"Input_size of MLP: {input_size}")

```

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Image shape: torch.Size([1, 28, 28])

Input_size of MLP: 784

```
In [3]: # select 16 random indices for plotting
indices = random.sample(range(len(train_dataset)), 16)

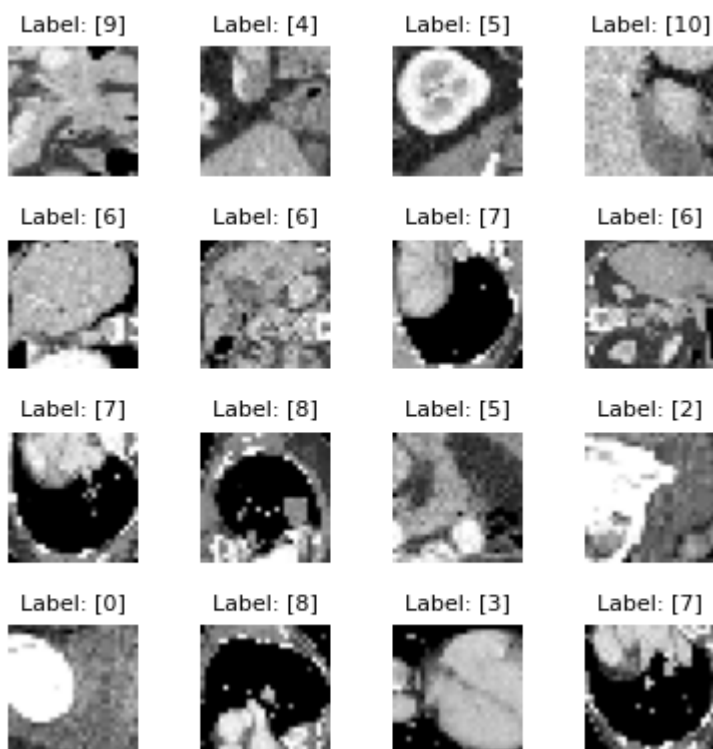
plt.figure(figsize=(4, 4))

for i, idx in enumerate(indices):
    image, label = train_dataset[idx]

    # If tensor, convert to 2D array (grayscale)
    if hasattr(image, 'squeeze'):
        image = image.squeeze() # remove channel dimension if present

    plt.subplot(4, 4, i + 1)
    plt.imshow(image, cmap='gray')
    plt.title(f"Label: {label}", fontsize=8)
    plt.axis('off')

plt.tight_layout()
plt.show()
```



```
In [4]: # Fully connected neural network with one hidden layer
class MLP(nn.Module):
    def __init__(self, input_size, hidden_size, num_classes):
        super(MLP, self).__init__()
        self.input_size = input_size
        self.l1 = nn.Linear(input_size, hidden_size)
        self.relu = nn.ReLU()
        self.l2 = nn.Linear(hidden_size, num_classes)

    def forward(self, x):
        out = self.l1(x)
        out = self.relu(out)
        out = self.l2(out)
        # no activation and no softmax at the end
        #return torch.Log_softmax(out, dim=1) #for NLL Los
        return out
```

```
mlp_model = MLP(input_size, hidden_size, n_classes).to(device)
# Loss and optimizer
criterion = nn.CrossEntropyLoss() # One can try NLL Loss
optimizer = torch.optim.Adam(mlp_model.parameters(), lr=lr)
```

```
In [5]: # Training
def train(model, dataloader, criterion, optimizer, device):
    model.train()
    train_correct = 0
    total_samples = 0

    for inputs, targets in dataloader:
        inputs, targets = inputs.reshape(-1, 28*28).to(device), torch.squeeze(targets)

        optimizer.zero_grad()
        outputs = model(inputs)
        loss = criterion(outputs, targets)
        loss.backward()
        optimizer.step()

        _, predictions = torch.max(outputs, 1)
        train_correct += (predictions == targets).sum().item()
        total_samples += targets.size(0)

    train_acc = (train_correct / total_samples) * 100
    return train_acc, loss.item()

# Validation/Test
def val(model, dataloader, criterion, device):
    model.eval()
    val_correct = 0
    total_samples = 0
    val_loss = 0.0

    with torch.no_grad():
        for inputs, targets in dataloader:
            inputs, targets = inputs.reshape(-1, 28*28).to(device), torch.squeeze(targets)

            outputs = model(inputs)
            loss = criterion(outputs, targets)
            _, predictions = torch.max(outputs, 1)

            val_correct += (predictions == targets).sum().item()
            total_samples += targets.size(0)
            val_loss += loss.item()

    val_acc = (val_correct / total_samples) * 100
    avg_val_loss = val_loss / len(dataloader)
    return val_acc, avg_val_loss
```

```
In [6]: Train_Acc = []
        Train_Loss = []

        Val_Acc = []
        Val_Loss = []

        for epoch in range(n_epochs):
```

```

train_acc, train_loss = train(mlp_model, train_loader, criterion, optimizer,
val_acc, val_loss = val(mlp_model, val_loader, criterion, device)
print (f'Epoch--[{epoch+1}], Train_Accuracy: {train_acc}, Validation_Accuracy: {val_acc}')
Train_Acc.append(train_acc)
Train_Loss.append(train_loss)
Val_Acc.append(val_acc)
Val_Loss.append(val_loss)

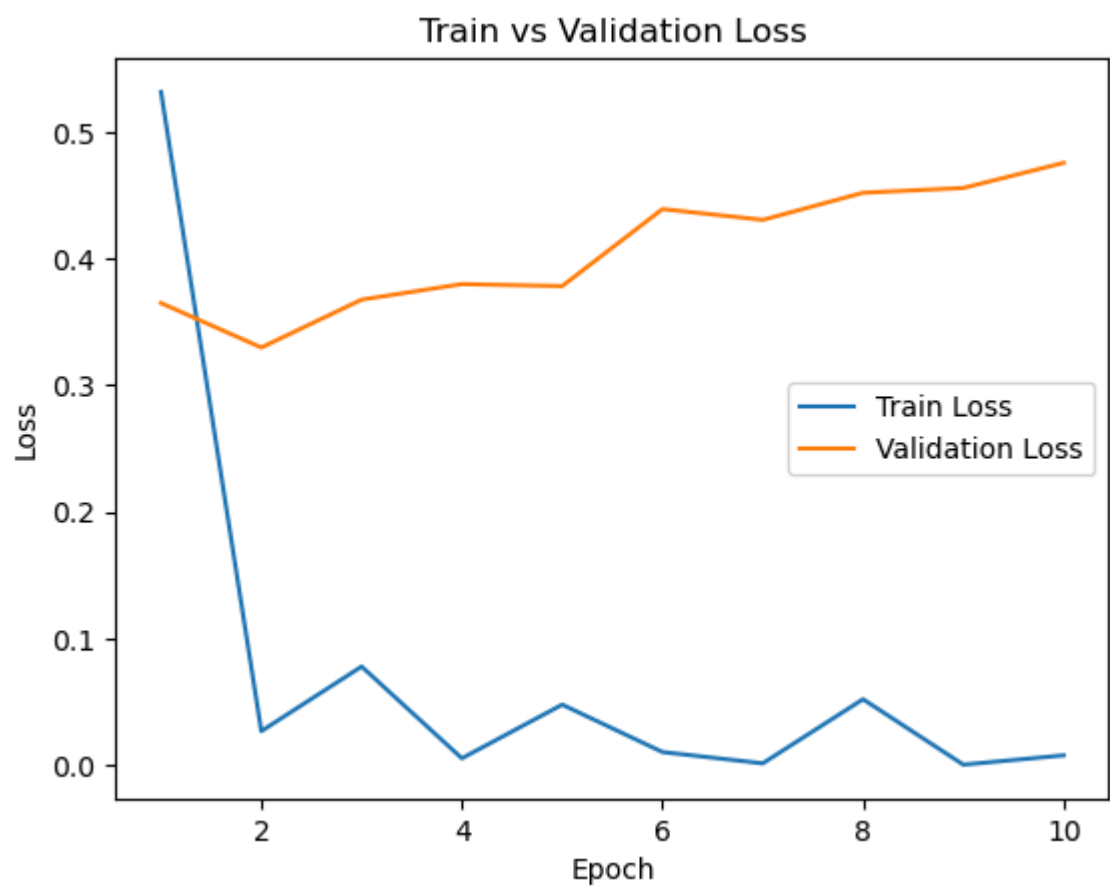
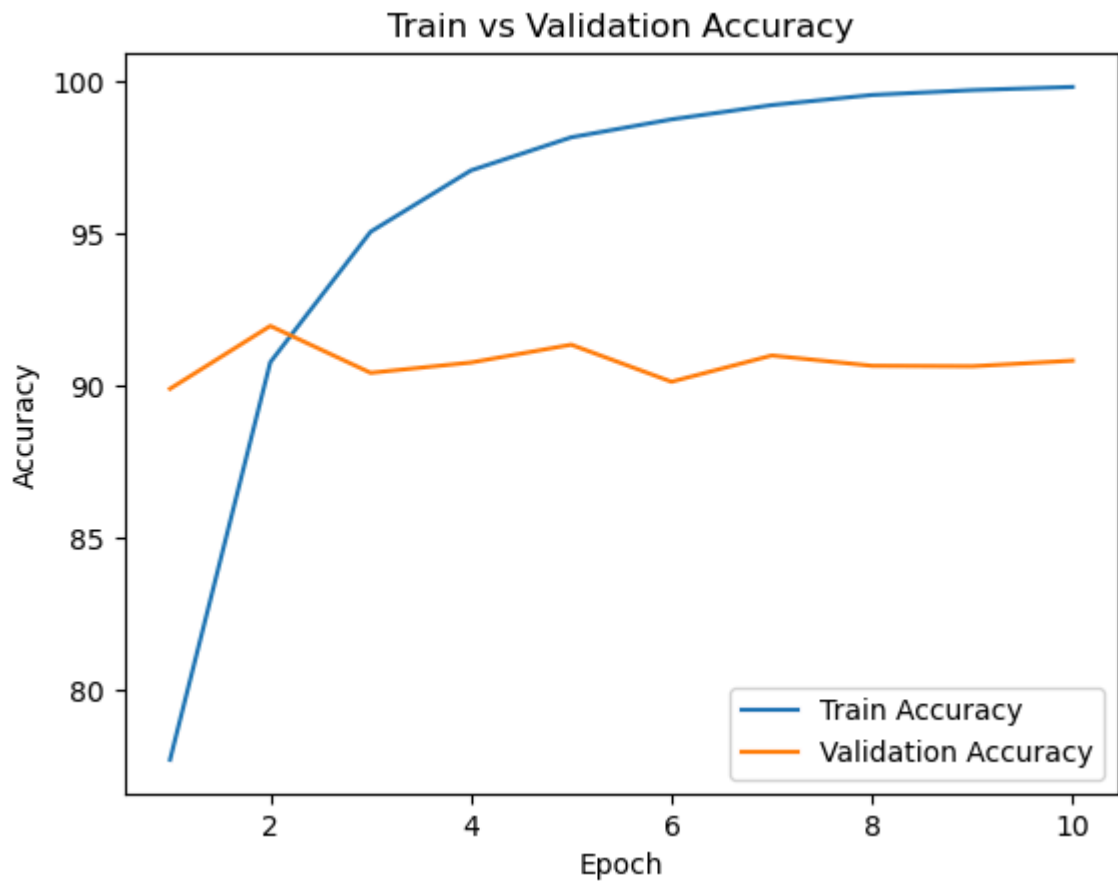
# Plot Epoch vs Accuracy
plt.figure()
plt.plot(range(1, n_epochs+1), Train_Acc, label='Train Accuracy')
plt.plot(range(1, n_epochs+1), Val_Acc, label='Validation Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.title('Train vs Validation Accuracy')
plt.legend()
#plt.grid(True)
plt.show()

# Plot Epoch vs Loss
plt.figure()
plt.plot(range(1, n_epochs+1), Train_Loss, label='Train Loss')
plt.plot(range(1, n_epochs+1), Val_Loss, label='Validation Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Train vs Validation Loss')
plt.legend()
#plt.grid(True)
plt.show()

#Testing
test_acc, test_loss = val(mlp_model, test_loader, criterion, device)
print (f'Test_Accuracy: {test_acc}')
```

```

Epoch--[1], Train_Accuracy: 77.70897832817337, Validation_Accuracy: 89.8936989678
0157
Epoch--[2], Train_Accuracy: 90.76704956453806, Validation_Accuracy: 91.9580958249
8845
Epoch--[3], Train_Accuracy: 95.05222649807587, Validation_Accuracy: 90.4175011554
46
Epoch--[4], Train_Accuracy: 97.06605711640289, Validation_Accuracy: 90.7564319827
4534
Epoch--[5], Train_Accuracy: 98.1539885998669, Validation_Accuracy: 91.34185795717
147
Epoch--[6], Train_Accuracy: 98.74424929834205, Validation_Accuracy: 90.1247881682
3294
Epoch--[7], Train_Accuracy: 99.21298573536646, Validation_Accuracy: 90.9875211831
7671
Epoch--[8], Train_Accuracy: 99.54573073695784, Validation_Accuracy: 90.6485903558
7737
Epoch--[9], Train_Accuracy: 99.70486965076242, Validation_Accuracy: 90.6331844091
8195
Epoch--[10], Train_Accuracy: 99.8090333034345, Validation_Accuracy: 90.8180557695
2704
```



Test_Accuracy: 73.79345258184273

In []: